METRIC

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w/AMENDMENT 2
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SUPERSEDING
MIL-DTL-47059A
w/AMENDMENT 1
02 September 2021

DETAIL SPECIFICATION

ACETYL TRIETHYL CITRATE

Reactivated after 08 Apr 21 and may be used for new and existing designs acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for acetyl triethyl citrate.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E203	-	Standard Test Method for Water Using Volumetric Karl Fischer Titration
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM E1064	-	Standard Test Method for Water in Organic Liquids by Coulometric Karl Fischer Titration.
ASTM D1193	-	Standard Specification for Reagent Water
ASTM D1209	-	Color of Clear Liquids (Platinum-Cobalt Scale)

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see paragraph 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Physical and chemical properties. Acetyl triethyl citrate shall conform to the physical and chemical requirements specified in Table I.

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TABLE I. Chemical and physical requirements for acetyl triethyl citrate.

Test	Characteristics
Assay (total ester), weight percent, minimum (min.)	99.00
Moisture, weight percent, maximum (max.)	0.30
Color (preferred method) APHA number, max Color (alternate method) light absorbance value for the highest peak within 440 to 550 nm, max	 50 10±1% of the measured APHA 500 unit stock solution value
Acidity (as citric acid), percent, max.	0.02

3.3 Workmanship. Acetyl triethyl citrate shall be uniform in quality, free from visible impurities and foreign material such as dirt, wooden particles or any other foreign matter so that the safety and reliability of the material are not compromised.

4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Assay	X		X	X	X	X	4.4.1
3.2	Moisture	X		X	X	X	X	4.4.2
3.2	Color	X		X	X	X	X	4.4.3
3.2	Acidity	X		X	X	X	X	4.4.4
3.3	Workmanship	X		X	X	X	X	4.4.5

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).

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b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see paragraph 6.2), a sample shall be subjected to first article verification in accordance with Table III. Samples of not less than six hundred (600) milliliters (mL) of acetyl triethyl citrate collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

TABLE III. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Assay	3.2	4.4.1
102	Moisture	3.2	4.4.2
103	Color	3.2	4.4.3
104	Acidity	3.2	4.4.4
Minor			
201	Workmanship	3.3	4.4.5

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements, the first article batch shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each batch shall consist of a quantity of acetyl triethyl citrate that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one acceptable batch of acetyl triethyl citrate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2. Acetyl triethyl citrate sampling.

4.3.2.1 Sampling at the manufacturer. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 600 mL shall be formed which is assumed to be representative from a single batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the lot shall be rejected.

4.3.2.2 Sampling at the received facility. Acetyl triethyl citrate batches shipped in containers (e.g., drums, totes, boxes, etc.) shall be sampled and accepted using Verification Level I of MIL-STD-1916, where one unit is defined as one container. Each container shall be sampled in three (3) locations of the shipping containers in accordance with ASTM E300. This material

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shall be placed in a secondary container and mixed to form a composite sample consisting of a minimum of 600 mL for each acetyl triethyl citrate lot received. There shall be no reduced sampling. Every lot received shall be tested and verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the lot shall be rejected.

4.3.3 Classification of characteristics. Acetyl triethyl citrate shall be subjected to the inspections according to 4.3.3.1.

4.3.3.1 Acetyl triethyl citrate.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Assay	4.3.2	3.2	4.4.1
102	Moisture	4.3.2	3.2	4.4.2
103	Color	4.3.2	3.2	4.4.3
104	Acidity	4.3.2	3.2	4.4.4
Minor				
201	Workmanship	4.3.2	3.3	4.4.5

4.4 Test methods and procedures.

4.4.1 Assay (as total ester).

4.4.1.1 Reagents.

a. Isopropyl alcohol, reagent grade
b. Sodium hydroxide, 0.5N
c. Standardized sulfuric acid, 0.5N
d. Phenolphthalein indicator

4.4.1.2 Procedure.

a. Accurately weigh 1.5 grams (g) of sample, to the nearest 0.001 g, into a 500 mL flask with standard taper ground-glass neck.
b. Add 25 mL of isopropyl alcohol and 25 mL of distilled water.
c. Pipette exactly 50 mL of 0.5N sodium hydroxide into the reaction mixture.
d. Add boiling chips.
e. Attach the flask to a water-cooled condenser with standard taper joint.
f. Reflux for not less than 1½ hours.
g. Cool and wash down the condenser with distilled water.
h. Add 5 drops of phenolphthalein indicator. Titrate with 0.5N sulfuric acid to a colorless endpoint.

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i. A blank using identical quantities of reagents shall be run simultaneously with the sample.

$$Percent\ Assay\ (as\ total\ ester) = \frac{(A - B)\ x\ N\ x\ 0.0796}{C}\ x\ 100$$

Where:

A = volume of sulfuric acid needed to titrate blank, mL
B = volume of sulfuric acid needed to titrate sample, mL
N = normality of sulfuric acid
C = weight of sample, g

4.4.2 Moisture.

4.4.2.1 Volumetric titration moisture determination (preferred method). Moisture content shall be determined by the Karl Fischer volumetric method using a direct-titration technique in accordance with ASTM E203.

4.4.2.1.1 Equipment.

a. Mettler Toledo V20S Karl Fischer titrator or equivalent (see paragraph 6.5). Equipped with a 5 mL burette (for preferred calibration) or 10 mL burette (for alternate calibration)
b. 3 mL or 5 mL syringe, 20 gauge or larger (lower gauge number) with 4 inch (10 cm) cannula or needle, for preferred calibration
c. 10 mL syringe, 20 gauge or larger (lower gauge number) with 4 inch (10 cm) cannula or needle, for alternate calibration
d. A 50 or 100 uL (microliter) gas tight syringe with an 18 to 24 gauge, 2 inch (5 cm) needle with blunt, conical or bevel tip, for alternate calibration

4.4.2.1.2 Reagents.

a. Methanol, reagent grade, containing less than 0.01% of water, in the titration vessel
b. Hydranal One Component Composite 2, nominal concentration of 2 milligrams (mg) water per mL titrant or equivalent (see 6.5)
c. Hydranal One Component Composite 5, nominal concentration of 5 milligrams (mg) water per mL titrant or equivalent (see 6.5)
d. De-ionized (DI) water (ASTM D1193, Type III), or distilled water (ASTM D1193, Type II or III) for alternate calibration, or disodium tartrate dihydrate standard, containing 15.66% water for preferred calibration

4.4.2.1.3 Instrument settings. (Recommended settings for Mettler Toledo V20S Karl Fischer titrator).

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a. Drift: 25 µg/minute (microgram/minute)
b. Sensor current (Ipol): 15.0 µA (microAmpere)
c. Stirrer speed: 30% (may be adjusted as needed, ensure that electrodes are fully immersed)
d. Stirrer time: 60 seconds

4.4.2.1.4 Titrant calibration using disodium tartrate dihydrate (preferred calibration).

a. Fill the titration vessel with approximately 40 mL methanol. Condition the solvent to the endpoint by titrating with Hydranal One Component Composite 2.
b. Weigh approximately 0.030 to 0.040 g of disodium tartrate dihydrate on a weighing funnel. Record the weight to the nearest 0.0001 g.
c. Set the instrument to perform concentration:
    I. Press the Start Concentration key.
    II. Remove the stopper from the titrator vessel and quickly transfer the standard solution into the vessel.
    III. Return the stopper to the vessel. The titration will initiate. Note: the start concentration key will not be active until the solvent is conditioned.
d. Weigh the transfer funnel. Record the weight to the nearest 0.0001 g.
e. Calculate the standard weight and enter the weight in grams into the titrator.
f. Upon completion of the titration, record the titer value.
g. The titer determination (steps b-f) shall be performed in triplicate. Do not replace the reagent/solvent for each determination.
h. Record the results as follows: mean, standard deviation and relative standard deviation (RSD). Note: the RSD shall not exceed 2.0%. If it does, repeat the titer determination after ensuring that the vessel seals are secure and the drying agents are active.

4.4.2.1.5 Sample moisture determination (preferred method).

a. Replace the solvent in the titration vessel with fresh methanol and condition the solvent to the endpoint. (see paragraph 4.4.2.1.4.a)
b. Draw approximately 3 mL of the sample into the syringe. Weigh the syringe in grams. Record the results to the nearest 0.0001 g.
c. Press the Start Sample key, then dispense the sample into the titration vessel with the needle or cannula tip slightly below the surface of the solvent. The vessel stopper should be kept open for the shortest possible period to minimize the introduction of atmospheric moisture into the vessel. The titration will begin when the instrument senses the added water. Note: the Start Sample key will not be active until the solvent is conditioned to the endpoint.

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d. Reweigh the syringe and calculate the sample weight. Enter the sample weight into the titrator.
e. When the titration is completed, the titrator will display the sample moisture content. If the titration volume exceeds 5 mL, reduce the sample volume to 2 mL.
f. The moisture determination shall be tested in triplicate. It is not necessary to refill the vessel with fresh methanol after each determination, unless the volume in the vessel exceeds 100 mL. If it is replaced, the methanol shall be conditioned to the endpoint prior to the sample introduction.
g. Record the results as follows: mean, standard deviation and relative standard deviation (RSD). Note: the RSD shall not exceed 2.0%. If it does, repeat the sample moisture determination after verifying the vessel seals are secure and the drying agents are active.

4.4.2.1.6 Titrant calibration using DI or distilled water (alternate calibration).

a. Fill the titration vessel with approximately 40 mL methanol. Condition the solvent to the endpoint by titrating with Hydranal One Component Composite 5.
b. Weigh approximately 0.040 g of DI or distilled water using a gas-tight syringe. Record the weight to the nearest 0.0001 g.
c. Set the instrument to perform concentration:
    I. Press the Start Concentration key.
    II. Remove the stopper from the titrator vessel and quickly transfer the water calibrant into the vessel.
    III. Return the stopper to the vessel. The titration will initiate. Note: the start concentration key will not be active until the solvent is conditioned.
d. Weigh the emptied syringe. Record the weight to the nearest 0.0001 g.
e. Calculate the standard weight and enter the weight in grams into the titrator.
f. Upon completion of the titration, record the titer value.
g. The titer determination shall be tested in triplicate. Do not replace the reagent/solvent for each determination.
h. Record the results as follows: mean, standard deviation and relative standard deviation (RSD). Note: the RSD shall not exceed 2.0%. If it does, repeat the titer determination after ensuring that the vessel seals are secure and the drying agents are active.

4.4.2.1.7 Sample moisture determination (alternate method).

a. Replace the solvent in the titration vessel with fresh methanol and condition the solvent to the endpoint. (see paragraph 4.4.2.1.6.a)
b. Draw approximately 10 mL of the sample into the syringe. Weigh the syringe in grams. Record the results to the nearest 0.0001 g.

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c. Press the Start Sample key, then dispense the sample into the titration vessel with the needle or cannula tip slightly below the surface of the solvent. The vessel stopper should be kept open for the shortest possible period to minimize the introduction of atmospheric moisture into the vessel. The titration will begin when the instrument senses the added water. Note: the Start Sample key will not be active until the solvent is conditioned to the endpoint.
d. Reweigh the syringe and calculate the sample weight. Enter the sample weight into the titrator.
e. When the titration is completed, the titrator will display the sample moisture content. If the titration volume exceeds 10 mL, reduce the sample volume to less than 10 mL.
f. The moisture determination shall be tested in triplicate. It is not necessary to refill the vessel with fresh methanol after each determination, unless the volume in the vessel exceeds 100 mL. If it is replaced, the methanol shall be conditioned to the endpoint prior to the sample introduction.
g. Record the results as follows: mean, standard deviation and relative standard deviation (RSD). Note: the RSD shall not exceed 2.0%. If it does, repeat the sample moisture determination after verifying the vessel seals are secure and the drying agents are active.

4.4.2.2 Coulometric titration moisture determination (alternative method). Moisture content shall be determined by the Karl Fischer method using coulometric direct-titration technique in accordance with ASTM E1064.

4.4.2.2.1 Equipment.

a. Mettler Toledo C30 Coulometric Karl Fischer titrator or equivalent (see paragraph 6.5). Equipped with a frit-less generating electrode.
b. 3 mL or 5 mL syringe, 20 gauge or larger (lower gauge number) with 4 inch cannula or needle

4.4.2.2.2 Reagent.

a. Apura Combi Coulomat frit-less, 100 mL in the titration vessel

4.4.2.2.3 Sample moisture determination preparation by direct sample addition.

a. Draw a 1.0 mL sample into the syringe. Weigh the syringe in grams. Record the weight to the nearest 0.0001 g.
b. Initiate the method and allow the solvent to condition to the end point.
c. Press the Start Sample key, then dispense the sample into the titration vessel with the needle or cannula tip slightly below the surface of the solvent. The vessel stopper should be kept open for the shortest possible period to minimize the introduction of atmospheric moisture into the vessel. The titration will start when the instrument

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senses the added water. The Start Sample key will not be active until the solvent is conditioned to the endpoint.

d. Reweigh the syringe. Record the weight to the nearest 0.0001 g.
e. Calculate the sample weight in grams. Enter the sample weight into the titrator.
f. When the titration is completed, record the sample moisture percent to the nearest 0.001%.
g. The moisture determination shall be tested in triplicate. Do not replace the reagent/solvent for each determination. (see note below)
h. Record the results as follows: mean, standard deviation and relative standard deviation (RSD). The relative standard deviation shall not exceed 2.0%. If it does, repeat the sample moisture determination after verifying the vessel seals are secure and the drying agents are active.

Note: The coulometric reagent may need to be replaced if any of these conditions applies: the reagent bottle was opened for more than 1 month, the reagent becomes exhausted during the course of multiple individual determinations, this will be evident if the titration is not complete or does not reach equilibrium point, or If the coulometric titrator has been unused for a prolonged period of time, e.g. a month. If so, empty the vessel then pour Apura Combi Coulomat frit-less solution into the vessel up to the 100 mL line. Allow the solution to condition to the endpoint and resume the analysis.

In addition, Coulometric titrators do not require calibration. However, proper function of the instrument may be checked through the addition of known amounts of water in the form of sodium tartrate dihydrate standard containing 15.66% water.

4.4.3 Color.

4.4.3.1 Platinum – Cobalt scale color determination (preferred method). Color shall be verified in accordance with ASTM D1209.

4.4.3.1.1 Reagents.

4.4.3.1.1.1 Reagent for Standard stock (APHA 500 units, certified) preparation.

a. 500 Units APHA color standard stock solution, certified
b. De-ionized (DI) water

4.4.3.1.1.2 Reagents alternate standard preparation.

a. Hydrochloric acid (HCl), concentrated, 36.5 – 38%, analytical reagent grade
b. Cobalt chloride, ACS reagent grade
c. Potassium chloroplatinate (IV) >99.99%

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4.4.3.1.2 Equipment.

a. 50 mL Nessler tubes with caps
b. 50 mL Nessler racks, capable of holding at least 10 tubes
c. Graduated cylinders/volumetric pipettes (500 mL to 5 mL)
d. Glass funnel
e. Glass stirring rod (if alternate reagents are used)
f. 1-Liter beaker (if alternate reagents are used)
g. 1-Liter volumetric flask (if alternate reagents are used)
h. Plastic weighing boats (if alternate reagents are used)
i. Weighing scale (if alternate reagents are used)

4.4.3.1.3 APHA color standards preparation.

a. Standard stock (APHA 500 units, certified)

    1. Mark APHA units on Nessler tubes in accordance with Table IV.
    2. Transfer aliquots of the stock APHA 500 unit solution to corresponding Nessler tubes, as listed in Table IV.
    3. Dilute each Nessler tube to volume (exactly 50 mL) with DI water.
    4. Cap the tubes. The caps may be sealed with parafilm as long as the top of cap is not covered or partially obstructed by the parafilm.
    5. Place the filled Nessler tubes (APHA color standards) in the Nessler tube racks so that every other slot is open.

b. Alternate standard preparation.

    1. Add 100 mL of HCl to 400 mL DI water in 1 L beaker.
    2. Accurately weigh 1.245 g of potassium chloroplatinate and add to HCl solution.
    3. Accurately weigh 1.0 g of cobalt chloride to the nearest 0.01g and add to HCl solution.
    4. Stir until dissolved.
    5. Transfer solution to a volumetric flask, dilute to volume with DI water, and mix well. This stock solution has an APHA color index of 500 units.
    6. Prepare Nessler tube APHA color standards in accordance with 4.4.3.1.3.a.

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TABLE IV. Standards.

Aliquot (mL)[1]	Standard (APHA units)
0.50	5
1.00	10
2.00	20
3.00	30
4.00	40
5.00	50
7.50	75
10.00	100
15.00	150
20.00	200
30.00	300
40.00	400
50.00	500
Note[1]: Standards aliquot solutions in Table IV are one-half (1/2) from ASTM D1209.	

4.4.3.1.4 Acetyl triethyl citrate color analysis.

a. Accurately fill an empty 50 mL Nessler tube to volume with acetyl triethyl citrate sample. Add cap. The cap may be sealed with parafilm as long as the top of cap is not covered or obscured by the parafilm.
b. Place the acetyl triethyl citrate sample in empty slot between the standard sample tubes in rack.
c. View the color from the top of the tube (through cap), and compare with the color in the tubes on either side. If the color does not match either tube, transfer tube to the next open slot and continue comparing colors until the closest match is found.
d. Record the APHA color unit that is the closest match to the test sample. Note: if the Nessler tube racks have canted reflective white bases, the tube colors may also be observed from the bottom of Nessler tube rack by placing the rack under a bright light source and observing the base from a suitable angle and distance.
e. Additional APHA standard tubes may be prepared if a more accurate color estimation is needed.

4.4.3.2 Color determination method by UV/VIS light absorption analysis (alternate method).

4.4.3.2.1 Reagents.

a. As identified in 4.4.3.1.1

4.4.3.2.2 Equipment.

a. UV/VIS spectrometer
b. 10 mm (1 cm) quartz cuvettes

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c. 100-ml beaker

4.4.3.2.3 APHA 50 unit stock sample preparation.

a. Prepare APHA 50 unit color standard by accurately transfer 5 mL of stock APHA 500 unit solution into 100 mL beaker.
b. Accurately add 45 mL of DI water to beaker, and mix well.

4.4.3.2.4 Light absorbance threshold verification.

a. Fill a clean cuvette with stock APHA 500 unit solution.
b. Place sample in UV/VIS, and perform scan.
c. Record the absorbance of the highest peak. The peak shall be located within the 440 to 500 nm wavelength region of the spectrum.
d. Repeat steps a, b & c twice, and calculate the average absorbance. Record the result to four decimal places.
e. Fill a clean cuvette with APHA 50 unit color standard solution, and repeat steps b, c, & d above. The average absorbance of the highest peak within 440 to 500 nm wavelength region shall be equivalent to 10±1% of the measured APHA 500-unit stock solution. If it is not, prepare another APHA 50-unit solution, and repeat this step. If a light absorbance value equal to 10±1% of the APHA 500-unit stock solution cannot be obtained, then perform the standard APHA color test method detailed in section 4.4.3.1.

4.4.3.2.5 Acetyl triethyl citrate light absorption determination.

a. Fill a clean cuvette with acetyl triethyl citrate sample.
b. Place sample in UV/VIS, and perform scan.
c. Record the absorbance of the highest peak located within the 440 to 500 nm wavelength spectrum region.
d. Repeat step b, and c for two additional acetyl triethyl citrate samples, and determine the average absorbance for all three samples.

4.4.4 Acidity (as citric acid).

4.4.4.1 Reagents

a. Neutralized isopropyl alcohol, reagent grade
b. Standardized sodium hydroxide, 0.1N
c. Bromothymol blue indicator, 0.04% aqueous solution

4.4.4.2 Procedure. Accurately weigh 30 to 50 g of acetyl triethyl citrate to the nearest 0.0001 g and dissolve it in 30 mL neutralized isopropyl alcohol. Add 5 drops of bromothymol blue indicator and titrate to neutrality with 0.1N sodium hydroxide. Calculate percent acidity as follows:

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$$Percent\ free\ acidity = \frac{A\ x\ N\ x\ 0.064}{B}\ x\ 100$$

Where: A = sodium hydroxide used in titration, mL
0.064 = milliequivalent for citric acid
B = weight of sample, g
N = normality of sodium hydroxide

4.4.5 Workmanship. Place approximately 100 mL of acetyl triethyl citrate into a 150 mL glass beaker over a clean, well-lighted white surface. The sample shall be examined by personnel with normal vision. The sample shall be uniform in quality and free from visible impurities, and foreign material such as dirt, wooden particles or any other foreign matter. Failure to meet the workmanship requirement during inspection or while executing production processes shall result in batch rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see paragraph 6.2). When packaging of the materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Acetyl triethyl citrate is intended to be used as an ingredient in ammonium-nitrate-based solid propellants and plasticized cellulose acetate restrictor material.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
c. Provisions for submission/resubmission of first article samples
d. Quantity required and delivery schedules
e. Levels of preservation and packing (see 5.1)
f. Item hazard classification
g. Certificate(s) of Analysis reporting the test results for the batch(es) or container(s) Tested.

6.3 Definitions.

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6.3.1 Batch. A discrete quantity of material that has been manufactured through various unit operations to produce a substantially uniform final product.

6.4 Technical agency. US Army Combat Capability Development Command Armament Center (DEVCOM AC) is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods, instruments and/or reagents. Prior approval of the Contracting Officer is required for use of equivalent test methods, instruments and/or reagents. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.4) responsible for preparation of this specification.

6.7 Incorporated changes. ECP Y21Q2004 provides documentation for noted changes.

6.8 Government limited liability. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.9 Changes incorporated. This specification was updated from MIL-A-47059 to MIL-DTL-47059 with Amendment 2 as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Packaging instructions have been changed. Sampling procedures have been updated. Details of these changes are documented in the ECPs and ERR listed below.

Specification ID	ECP / ERR
MIL-A-47059	-
MIL-DTL-47059A	Y20Q2009
MIL-DTL-47059A w/ Amendment 1	Y21Q2004
MIL-DTL-47059A w/ Amendment 2	Y24Q3009

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6.10 Subject term (key word) listing.

Energetic material
Propellant
Ingredient
Plasticizer
Munitions Chemical Specification Modernization (MCSM)

6.10 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 70

Preparing activity
Army - AR
(Project: 6810-2025-010)

Review Activities:
Air Force - 170

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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